SOLUTION MANUAL

PART I: TEMPERATURE, HEAT AND THE FIRST LAW OF THERMODYNAMICS

Exercise 1.

The change in length for the section of the steel ruler between its 20.05 cm mark and 20.11 cm mark is

$\Delta L_s = L_s \Delta T$ = 20.11 cm x 11x10-6/°C x (270°C-20°C) = 0.055 cm.

Thus, the actual change in length for the rod is

ΔL_r = (20.11 cm – 20.05 cm) + 0.055 cm = 0.115 cm

The coefficient of thermal expansion for the material of which the rod is made is then

$\alpha = \frac{L}{TLr}$ = 23x10^{-6} /C^{o}

Exercise 2.

(a) The specific heat is given by $c = Q/m(T_f - T_i)$, where Q is the heat added, m is the mass of the sample, T_i is the initial temperature, and T_f is the final temperature. Thus, recalling that a change in Celsius degrees is equal to the corresponding change on the Kelvin scale,
c = 523 J/kg.K
(b) The molar specific heat is given by
$c_m = Q / [N(T_f - T_i)]$ = 26.2 J/mol.K
(c) N = m/M = *30g/50 gmol^{-1} = 0.6 mol*

Exercise 3.
Let the mass of the steam be m_s and that of the ice be m_i. Then

|Qlost (steam)|= |Qreceived (ice)|

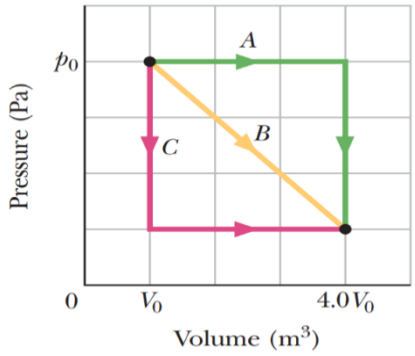
Qlost + Qreceived = 0

- $m_s L_s + m_s c_w (T_f - 100\ ^{o}C) + L_F m_i + c_w m_i (T_f - 0\ ^{o}C) = 0$

where Tf = 50°C is the final temperature. We solve for m_s:

ms = 33g

Exercise 4.



(a) One part of path *A* represents a constant pressure process. The volume changes from 1.0 m3 to 4.0 m3 while the pressure remains at 40 Pa. The work done is:

$$W_A = p\Delta V = (40\text{Pa})\ (4.0\text{m}^3 - 1.0\text{m}^3) = 1.2\text{x}10^2\ \text{J}$$

(b) The other part of the path represents a constant volume process. No work is done during this process. The total work done over the entire path is 120J. To find the work done over path B we need to know the pressure as a function of volume. Then, we can evaluate the integral $W = +pdV$. According to the graph, the pressure is a linear function of the volume, so we may write $p = a + bV$, where a and b are constants. In order for the pressure to be 40Pa when the volume is 1.0m^3 and 10Pa when the volume is 4.00m^3 the values of the constants must be $a = 50\text{Pa}$ and $b = -10\text{Pa/m}^3$. Thus,

$$p = 50\ \text{Pa} - (10\ \text{Pa/m}^3)V$$

and

$$W_B = \int_1^4 pdV = \int_1^4 (50 - 10V)dV = 200\text{J} - 50\text{J} - 80\text{J} + 5.0\text{J} = 75\text{J}$$

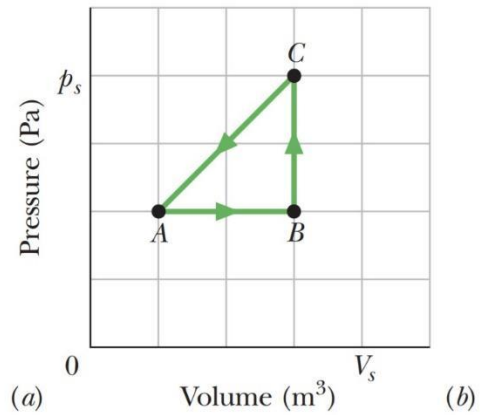
(c) One part of path *C* represents a constant pressure process in which the volume changes from $1.0\ \text{m}^3$ to $4.0\ \text{m}^3$ while p remains at 10 Pa. The work done is

$$W_C = p\Delta V = (10\text{Pa})(4.0\text{m}^3 - 1.0\ \text{m}^3) = 30\text{J}$$

The other part of the process is at constant volume and no work is done. The total work is

30 J. We note that the work is different for different paths.

Exercise 5.



(a) (b)

	Q	W	ΔE_{int}
$A \rightarrow B$	(a)	(b)	+
$B \rightarrow C$	+	(c)	(d)
$C \rightarrow A$	(e)	(f)	(g)

For a closed cycle: $\Delta E_{int} = 0$

During process $A \rightarrow B$, the system is expanding, doing work on its environment, so $W > 0$, and since $\Delta E_{int} > 0$ is given then $Q = W + \Delta E_{int}$ must also be positive.

(a) $Q > 0$.

(b) $W > 0$.

During process $B - C$, the system is neither expanding nor contracting. Thus,

(c) $W = 0$.

(d) The sign of ΔE_{int} must be the same (by the first law of thermodynamics) as that of Q,

which is given as positive. Thus, $\Delta E_{int} > 0$.

During process $C - A$, the system is contracting. The environment is doing work on the system, which implies $W < 0$. Also, $\Delta E_{int} < 0$ because $\Delta E_{int} = 0$ (for the whole cycle) and the other values of ΔE_{int} (for the other processes) were positive. Therefore, $Q = W + \Delta E_{int}$ must also be negative.

(e) $Q < 0$.

(f) $W < 0$.

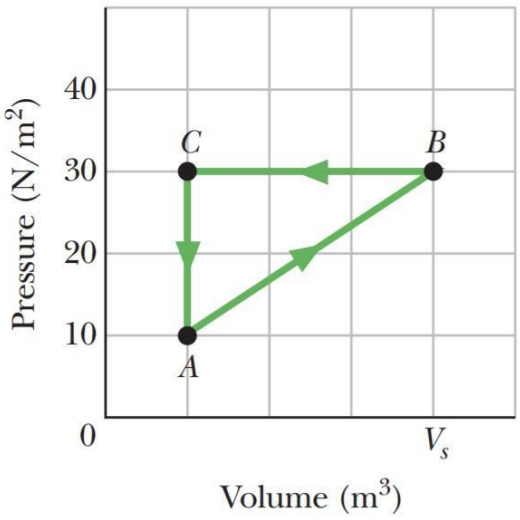
(g) $\Delta E_{int} < 0$.

(h) The area of a triangle is $\frac{1}{2}$ (base)(height). Applying this to the figure, we find

$|W_{net}| = \frac{1}{2}(2.0\ m^3)(20\ Pa) = 20\ J$.

Since process $C - A$ involves larger negative work (it occurs at higher average pressure) than the positive work done during process $A - B$, then the net work done during the cycle must be negative. The answer is therefore $W_{net} = -20$ J.

Exercise 6.



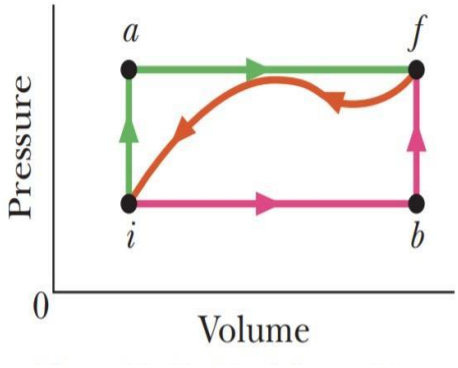
Over a complete cycle, the internal energy is the same at the beginning and end, $\Delta E_{int} = 0$, so the heat Q absorbed equals the work done: $Q = W$.

The *CA* portion of the cycle is at constant volume, so no work is done. The total work done by the gas is

$$W = W_{AB} + W_{BC} + W_{CA}$$

Over the portion of the cycle from *A* to *B* the pressure p is a linear function of the volume *V* and we may write $p = a + bV$. The work done over this portion of the cycle is

$$W_{AB} = \int_{VA}^{VB} pdV = \int_{VA}^{VB} (a + bV)dV = a(V_B - V_A) + \frac{1}{2} b(V_B^2 - V_A^2).$$



The *BC* portion of the cycle is at constant pressure and the work done by the gas is

$$W_{BC} = p_B \Delta V_{BC} = p_B(V_C - V_B)$$

The pressure function can be written as

$$p = \frac{10}{3}\ \text{Pa} + \left(\frac{20}{3}\ \text{Pa/m}^3\right)V,$$

where the coefficients *a* and *b* were chosen so that $p = 10$ Pa when $V = 1.0\ \text{m}^3$ and $p = 30$ Pa when $V = 4.0\ \text{m}^3$. Therefore, the work done going from *A* to *B* is

$$W_{AB} = a(V_B - V_A) + \frac{1}{2} b(V_B^2 - V_A^2) = 10\text{J} + 50\text{J} = 60\text{J}$$

Similarly, with $p_B = p_C = 30$ Pa, $V_C = 1.0\ \text{m}^3$ and $V_B = 4.0\ \text{m}^3$, we have

$$W_{BC} = p_B(V_C - V_B) = -90\ \text{J}.$$

Adding up all contributions, we find the total work done by the gas to be

$$W = W_{AB} + W_{BC} + W_{CA} = 60\ \text{J} - 90\ \text{J} + 0 = -30\ \text{J}$$

Thus, the total heat absorbed is $Q = W = -30$ J. This means the gas loses 30 J of energy in the form of heat.

Exercise 7.

(a) Since work is done on the system (perhaps to compress it) we write $W = -200$ J.

(b) Since heat leaves the system, we have $Q = -70.0$ cal $= -293$ J.

(c) The change in internal energy is $\Delta E_{\text{int}} = Q - W = -293\ \text{J} - (-200\ \text{J}) = -93$ J.

Exercise 8.

Since the change in internal energy ΔE_{int} only depends on the initial and final states, it is the same for path *iaf* and path *ibf*.

According to the first law of thermodynamics, $\Delta E_{\text{int}} = Q - W$, where *Q* is the heat absorbed and *W* is the work done by the system. Along *iaf*, we have

$$\Delta E_{\text{int}} = Q - W = 50\ \text{cal} - 20\ \text{cal} = 30\ \text{cal}.$$

(a) The work done along path *ibf* is given by

$$W = Q - \Delta E_{\text{int}} = 36\ \text{cal} - 30\ \text{cal} = 6.0\ \text{cal}.$$

(b) Since the curved path is traversed from *f* to *i* the change in internal energy is $\Delta E_{\text{int}} = -30$ cal, and

$$Q = \Delta E_{\text{int}} + W = -30 \text{ cal} - 13 \text{ cal} = -43 \text{ cal}.$$

(c) Let $\Delta E_{\text{int}} = E_{\text{int},f} - E_{\text{int},i}$. We then have

$$E_{\text{int},f} = \Delta E_{\text{int}} + E_{\text{int},i} = 30 \text{ cal} + 10 \text{ cal} = 40 \text{ cal}.$$

(d) The work W_{bf} for the path *bf* is zero, so

$$Q_{bf} = E_{\text{int},f} - E_{\text{int},b} = 40 \text{ cal} - 22 \text{ cal} = 18 \text{ cal}.$$

(e) For the path *ibf*, $Q = 36$ cal so $Q_{ib} = Q - Q_{bf} = 36 \text{ cal} - 18 \text{ cal} = 18 \text{ cal}$.

Exercise 9.

Since the process is a complete cycle (beginning and ending in the same thermodynamic state) the change in the internal energy is zero, and the heat absorbed by the gas is equal to the work done by the gas: $Q = W$. In terms of the contributions of the individual parts of the cycle $Q_{AB} + Q_{BC} + Q_{CA} = W$ and

$$Q_{CA} = W - Q_{AB} - Q_{BC} = 15.0 \text{ J} - 20.0 \text{ J} - 0 = -5.0 \text{ J}.$$

This means 5.0 J of energy leaves the gas in the form of heat.

Exercise 10.

We note that there is no work done in the process going from d to a, so $Q_{da} = \Delta E_{\text{int } da} = 80$ J. Also, since the total change in internal energy around the cycle is zero, then

$$\Delta E_{\text{int } ac} + \Delta E_{\text{int } cd} + \Delta E_{\text{int } da} = 0$$

$$-200 \text{ J} + \Delta E_{\text{int } cd} + 80 \text{ J} \quad = 0$$

which yields $\otimes E_{\text{int } cd} = 120$ J. Thus, applying the first law of thermodynamics to the *c* to *d* process gives the work done as

$$W_{cd} = Q_{cd} - \Delta E_{\text{int } cd} = 180 \text{ J} - 120 \text{ J} = 60 \text{ J}.$$

Exercise 11.

(a) We note that process a to b is an expansion, so $W > 0$ for it. Thus, $W_{ab} = 5.0$ J. We are told that the change in internal energy during that process is 3.0 J, so application of the first law of thermodynamics for that process immediately yields $Q_{ab} = 8.0$ J.

(b) The net work (1.2 J) is the same as the net heat ($Q_{ab} + Q_{bc} + Q_{ca}$), and we are told that $Q_{ca} = 2.5$ J. Thus, we readily find $Q_{bc} = (1.2 - 8.0 - 2.5)$ J = -9.3 J.

MCQs:

1. A
2. C
3. A
4. A
5. C

6. A
7. A
8. D
9. A
10. D